

Partition-Resolved Coherence in Quantum Algorithms

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Quantum computation does not provide branchwise readout of many classical evaluations. Its operational resource is coherent processing that encodes problem structure into amplitudes and resolves a selected property by interference. We formulate a decision problem as a partition of an instance set. Exact resolution is possible if and only if the encoded states associated with different partition cells occupy orthogonal output subspaces; the readout may then be represented either by a Hermitian observable whose spectral projectors label the cells or by a unitary that writes the cell label into an output register. A separating family of such partitions can support complete instance identification, although one partition measurement need not reveal the complete instance. As a concrete model, we introduce an inaccessible witness into the Bernstein–Vazirani oracle. Perfect witness records erase selected interference coordinates, while nonorthogonal witness states give a continuous visibility parameter and an explicit output distribution. The model separates global reversible evaluation from accessible, partition-resolving recombination and clarifies the roles of oracle access, erasure, contextuality, and uniformity.

I. INTRODUCTION

A common description of quantum computation says that a quantum computer evaluates a function on many inputs “at once.” The image is seductive: a register is placed in a superposition, a function is applied to all components, and a final measurement extracts the desired answer. This picture is often invoked when discussing Deutsch–Jozsa [1], Simon’s algorithm [2], Shor’s algorithm [3], Grover search [4], and related oracle or Fourier-sampling procedures [5, 6].

Taken literally, however, the picture is wrong. A quantum computer does not return a list of all evaluated values. The components of a superposition are not independently readable, copyable, or archivable classical records. They are amplitudes in one Hilbert-space state, and a run of the final measurement returns one outcome label—possibly a multibit label—with probabilities fixed by the Born rule.

The organizing thesis of this article is that many quantum decision and identification tasks are naturally encoded by a partition Π of a problem-instance set Ω . The immediate task is to identify the cell $[\omega]_\Pi$, rather than necessarily the complete instance ω . A suitable coherent encoding and recombination can make different cells orthogonal and hence exactly distinguishable. Equivalently, the cell may be resolved by a Hermitian operator whose eigenspaces represent the cells, or by a unitary operator that transfers the cell label to a readout register. A family of partitions whose common refinement is the singleton partition can, when the associated measurements are physically available, yield complete instance identification. This partition/operator formulation is developed in Sec. III.

This formulation does not make coarseness the source of quantum advantage. The operative requirement is an

end-to-end efficient match among state preparation, oracle access, coherent transformation, and readout. A measured partition label may also be an intermediate invariant from which classical post-processing recovers a complete answer, as in order finding and factoring. Conversely, an arbitrary partition of black-box data need not admit an efficient quantum resolution.

The claim is not meant to settle interpretive questions about Everettian or many-worlds accounts of quantum computation. Advocates of such accounts have argued that quantum algorithms can be understood in terms of computations distributed across branches or worlds [7–9]. The narrower claim is operational and model-theoretic: superposition components are not independently readable classical records, whereas coherent global evolution can convert phase relations into an accessible partition label or other task-relevant statistic.

The technical contribution is the leakage construction in Appendix A. It treats the accessible oracle as a channel obtained from a globally unitary interaction with an inaccessible witness. Besides the discrete case in which k path coordinates are recorded perfectly, a continuous version assigns nonorthogonal witness states with overlaps λ_i . This yields an explicit interpolation between exact Bernstein–Vazirani interference and ordinary value-query behavior. The later sections separately delimit what follows from unitary reversibility, Kochen–Specker contextuality, and computational uniformity.

II. SUPERPOSITION IS NOT BRANCHWISE READOUT

Let an n -qubit state be written in the computational basis as

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle.$$

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The dimension of the Hilbert space is 2^n , and this exponential dimension is often used to motivate quantum parallelism. But the amplitudes α_x are not classical memory cells. A measurement yields one outcome per run, with probabilities determined by the Born rule. Recovering the amplitudes requires many similarly prepared systems and many measurement settings, that is, tomography.

Thus a superposition can be acted on linearly, but it cannot be read branch by branch. Holevo's theorem gives one precise expression of this limitation. For an ensemble of states on a Hilbert space $\mathcal{H}$, the accessible classical information is bounded above by the Holevo quantity χ , and $\chi \leq \log_2 \dim \mathcal{H}$. Hence, for an unassisted ensemble encoded in an n -qubit system, the accessible classical information per use is at most n bits [6, 10]. (Entanglement-assisted communication is a different resource setting.)

The state vector may contain exponentially many amplitudes, but the accessible classical readout is sharply constrained. This is the first reason why the naive parallelism metaphor is misleading. The superposed components can interfere, but they are not simultaneously available classical facts.

III. PARTITIONS, COMPLETE IDENTIFICATION, AND RESOLVING OPERATORS

Let Ω be a finite set of problem instances. A decision or identification task is specified by a partition

$$\Pi = \{C_1, \dots, C_m\}, \quad \Omega = \bigsqcup_{j=1}^m C_j. \quad (1)$$

Here every C_j is nonempty, the cells are pairwise disjoint, and their union is Ω . Equivalently, Π is the set of equivalence classes of the relation $\omega \sim_\Pi \omega'$ if and only if ω and ω' belong to the same cell C_j . For an unknown instance $\omega \in \Omega$, the required answer is the cell $[\omega]_\Pi = C_j$ containing it. The task need not identify ω completely. Conversely, a family of partitions $\Pi^{(1)}, \dots, \Pi^{(r)}$ is *separating* when its common refinement is the partition whose cells are all nonempty intersections of one cell from each $\Pi^{(\ell)}$. We denote this common refinement by $\bigwedge_{\ell=1}^r \Pi^{(\ell)}$; it is the discrete partition when

$$\bigwedge_{\ell=1}^r \Pi^{(\ell)} = \{\{\omega\} : \omega \in \Omega\}. \quad (2)$$

Resolving all members of a separating family therefore gives complete instance identification. Here “complete identification” refers to the classical instance label; it is not tomography of an arbitrary unknown quantum state.

Suppose that the preparation, oracle call, and coherent post-processing associate an output state ρ_ω on a Hilbert space $\mathcal{H}_{\text{out}}$ with each instance. The partition Π

is resolved exactly by projectors $P_1, \dots, P_m$ when

$$P_j P_{j'} = \delta_{jj'} P_j, \quad \sum_{j=1}^m P_j = I, \quad (3)$$

$$\text{Tr}(P_j \rho_\omega) = 1 \quad \text{for } \omega \in C_j.$$

Thus the supports of states belonging to different cells must occupy orthogonal output subspaces. States within one cell need not be mutually distinguishable: their deliberately retained equivalence is precisely the coarse-graining encoded by Π .

For bounded-error algorithms, the exact projective condition is replaced by a POVM $\{M_j\}$ satisfying

$$M_j \succeq 0, \quad \sum_j M_j = I, \quad (4)$$

$$\text{Tr}(M_j \rho_\omega) \geq 1 - \epsilon \quad (\omega \in C_j), \quad \epsilon < \frac{1}{2}.$$

Different cells then need not have orthogonal supports; the relevant constraints are quantitative distinguishability bounds. The exact formulation above isolates the ideal case in which the desired partition label is perfectly resolvable and is not intended as a characterization of all bounded-error or sampling-based quantum algorithms.

Choose distinct real numbers $\lambda_1, \dots, \lambda_m$. The Hermitian operator

$$A_\Pi = \sum_{j=1}^m \lambda_j P_j \quad (5)$$

then resolves the partition: its outcome λ_j identifies C_j . Conversely, the spectral projectors of any Hermitian operator that exactly separates the cells supply projectors satisfying Eq. (3). A partition problem can therefore be encoded into, and resolved by, the spectral structure of a suitable observable.

There is an equivalent unitary form for pure encoded states. If $\mathcal{H}_j = \text{span}\{|\psi_\omega\rangle : \omega \in C_j\}$ and the subspaces $\mathcal{H}_j$ are mutually orthogonal, then an isometry, and hence a unitary extension on a sufficiently large space, can be chosen such that

$$U_\Pi |\psi_\omega\rangle |0\rangle_L = |\eta_\omega\rangle |j\rangle_L, \quad \omega \in C_j. \quad (6)$$

Measuring the label register L resolves the same partition. The Hermitian and unitary descriptions are not competing mechanisms: U_Π coherently transfers the cell information to a standard readout, and A_Π describes the associated spectral measurement. Equation (6) gives the corresponding label-register realization.

Complete identification by several partitions requires additional physical conditions. If the corresponding observables are jointly measurable, their joint spectral refinement may be measured directly. If they are incompatible, resolving all partitions generally requires separately prepared systems, sequential instruments with specified disturbance, or a different unitary that resolves the singleton refinement in one enlarged output space. Partition

separation is therefore a logical condition; the existence and efficiency of the required preparation, oracle, unitary, and measurement are the physical and computational conditions.

IV. THE USEFUL MECHANISM: STRUCTURE-MATCHED RECOMBINATION

Many oracle-based and Fourier-sampling quantum algorithms have the schematic form

$$\begin{aligned} |0\rangle^{\otimes n} &\xrightarrow{\text{spread}} \frac{1}{\sqrt{2^n}} \sum_x |x\rangle \\ &\xrightarrow{\text{compute}} \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \\ &\xrightarrow{\text{detect}} \text{measurable interference pattern.} \end{aligned}$$

The middle step may be produced by phase kickback. If

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

and the target register $|y\rangle$ is prepared in $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, then

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

The values of $f(x)$ are not read out individually. They are converted into phases. The final measurement is useful only if those phases are recombined in a basis matched to the structure one wants to detect.

This is the mechanism sometimes summarized as spread, compute, detect, or spread, encode, resolve. The algorithm distributes amplitude, encodes structure coherently, and recombines the amplitudes so that constructive and destructive interference concentrate a relevant invariant in the spectral subspaces of a chosen readout observable.

Here “structure-matched” is an operational criterion, not a claim that every successful algorithm has been explained merely by renaming its output. For a partition Π , the match consists of efficiently preparable inputs and oracle access, an efficiently implementable transformation that produces the orthogonal cell subspaces of Eq. (3), and an efficient readout of the corresponding observable. In the examples below the match takes two main forms: character, period, or hidden-subgroup structure, as in Bernstein–Vazirani, Simon, and Shor, where a Fourier transform turns an invariant into spectral concentration; and geometric concentration in a low-dimensional invariant subspace, as in Grover search and the Hein–Tanner lattice examples.

Deutsch–Jozsa is the simplest example. After phase kickback, the query register is in the state

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle.$$

Applying the final Hadamard transform $H^{\otimes n}$ gives the amplitude of $|0^n\rangle$ as

$$\langle 0^n | H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \right) = \frac{1}{2^n} \sum_x (-1)^{f(x)}.$$

This amplitude is ± 1 for constant functions and 0 for balanced functions. The promise makes the global predicate visible as complete constructive or destructive interference. The Deutsch–Jozsa promise problem therefore uses one exact quantum query, whereas a deterministic classical black-box algorithm requires $2^{n-1} + 1$ queries in the worst case. It is used here as a transparent interference example, not as a claim of an exponential bounded-error separation: randomized classical sampling distinguishes constant from balanced with $O(1)$ queries at bounded error.

This example also clarifies what the one-query quantum result concerns. It is not the recovery of a unique classical table of values. It is an exact solution of a decision problem whose answer is a coarse-grained class, because that class is exactly the invariant concentrated by the final interference pattern. For the original one-bit Deutsch problem, the four functions

$$f : \{0, 1\} \rightarrow \{0, 1\}$$

are not individually identified. They are partitioned into the two classes

$$\Pi_D = \{\{00, 11\}, \{01, 10\}\},$$

where the strings denote the ordered value pairs $f(0)f(1)$. Equivalently, the accessible answer is the parity

$$f(0) \oplus f(1),$$

or, in the usual terminology, the distinction between constant and balanced functions. The final interference pattern is therefore not a readout of $f(0)$ and $f(1)$ separately. It is a readout of the partition cell containing f .

In the operator language of Sec. III, the final Hadamard transform maps the two cells of Π_D to orthogonal output subspaces. Measuring the associated two-eigenvalue Hermitian observable resolves that partition, while a corresponding unitary may write the parity label to a separate register. Other partitions of the four truth tables can be combined with Π_D ; if their common refinement is discrete, their labels support complete truth-table identification, subject to the required oracle calls and compatibility conditions. Earlier work on quantum state identification used precisely this language for generalized Deutsch-type problems: trivalent functions of a trit were grouped into equipartitions, and the question was whether the corresponding functional classes could be separated by quantum state identifications [11].

This is not a strict opposition between partial and full answers. Coarse-graining is a common way for a task to

align with interference, but it is not the operative criterion. The operative criterion is the efficient physical realization of the partition projectors or their unitary label map. Arbitrary partitions of arbitrary black-box data need not have this property, and a partition-resolving intermediate statistic may sometimes be classically processed into a complete answer.

Shor’s algorithm is a more sophisticated example. Modular exponentiation is computed reversibly, but the algorithm does not read all values of

$$f(x) = a^x \bmod N.$$

Instead, periodicity is sampled through a quantum Fourier transform. The success comes from algebraic structure: translation symmetry becomes spectral concentration. Shor’s algorithm is therefore an important warning against saying that interference is useful only for “partial” knowledge. The quantum subroutine produces samples from which the order can be inferred; with the standard random choice of a , checks, and repetitions, continued-fraction and greatest-common-divisor post-processing yields a nontrivial factor of N with high probability. The final classical answer may be complete even when the quantum readout is an invariant.

The same observation explains why generic predicates are not made easy by superposition. Let a black-box string $b = (b_1, \dots, b_N)$ be accessed by the value oracle

$$O_b |i\rangle |y\rangle = |i\rangle |y \oplus b_i\rangle.$$

Its parity defines the two-cell partition

$$\begin{aligned} \Pi_{\text{par}} &= \{C_0, C_1\}, \\ C_p &= \left\{ b \in \{0, 1\}^N : \bigoplus_{i=1}^N b_i = p \right\}. \end{aligned}$$

Although Π_{par} has only two cells, they are maximally interlaced on the Boolean cube: flipping any one bit moves an instance from one cell to the other. Moreover, the projection of either cell onto any proper subset of the coordinates is the full cube on that subset. Thus every assignment to fewer than N coordinates has completions in both cells; no such partial coordinate information favors either parity label.

On the formal instance space spanned by the basis states $|b\rangle$, the corresponding partition observable may be written as

$$A_{\text{par}} = \sum_{b \in \{0, 1\}^N} (-1)^{\sum_i b_i} |b\rangle \langle b| = Z_1 Z_2 \cdots Z_N.$$

It is a weight- N observable: its eigenvalue is precisely the parity-cell label. This expression does not make the black-box problem trivial. If an accessible register already contained $|b\rangle$, the observable could be measured directly. Query complexity instead asks how many uses of O_b are required to encode this global label into distinguishable output subspaces. The partition must therefore be considered together with its oracle representation, not merely by counting its cells.

The polynomial method makes this obstruction precise. After T oracle queries, every output amplitude is a multilinear polynomial in the input bits of degree at most T , and every measurement probability has degree at most $2T$. Any probability that separates C_0 from C_1 with bounded error must approximate the parity character

$$(-1)^{\sum_i b_i} = \prod_{i=1}^N (-1)^{b_i},$$

whose approximate degree is N . Hence $2T \geq N$. This bound is tight: pairing the input positions allows the parity of each pair to be resolved with one quantum query, so exact parity requires $\lceil N/2 \rceil$ queries. The improvement over the deterministic classical query complexity is therefore at most a factor of two [12, 13].

The original Deutsch problem discussed above is precisely the $N = 2$ special case of this partition: $\Pi_D = \Pi_{\text{par}}$. Its two truth-table classes are already interlaced on the Boolean square, yet the bound $2T \geq N$ then requires only $T \geq 1$, which the one-query Deutsch algorithm attains. Interlacing is therefore compatible with a factor-of-two quantum reduction; for unrestricted parity it obstructs any larger asymptotic improvement [7, 12].

More generally, in the black-box query model, polynomial-method bounds show that for total Boolean functions the bounded-error quantum and classical query complexities are polynomially related, ruling out arbitrary exponential speedup for such functions [13]. Superposition alone is therefore not the resource. Structure-matched recombination is.

Appendix A gives a concrete interpolation for the Bernstein–Vazirani problem. A globally unitary oracle couples a tunable part of the computational-basis path label to an inaccessible witness. The Boolean function in the answer register is unchanged, but the oracle available to the algorithm is a reduced channel whose interference pattern is progressively suppressed. The appendix also shows that interference and full value recovery are composable: the surviving recombination recovers the coordinates still represented by orthogonal partition subspaces, and ordinary value queries can supply the rest.

Another illustration of this principle is provided by spatial-search and wave-communication constructions on regular lattices. The basic image is that of a coherent wave spread over a whole lattice. A locally marked site acts like a resonant defect: it is different enough from the background to be distinguished, but close enough in frequency to the extended lattice mode to exchange amplitude with it. The lattice then functions as a global carrier. In the lattice-search analysis of Hein and Tanner, this picture is made precise by an avoided crossing between the extended symmetric state and a localized defect state. The resulting dynamics rotates amplitude from the delocalized state into the marked localization. In their related wave-communication proposal, the same mechanism is extended to several marked vertices: with

two marked defects, amplitude can be transferred between sender and receiver through the symmetric carrier state [14, 15].

This is a spectral, spatial-search analogue of Grover-type amplitude amplification: coherent wave interference converts a phase or scattering perturbation into an observable amplitude concentration. Grover search also clarifies why “structure” should not be inflated until it explains everything. The database has no exploitable character, period, or hidden-subgroup promise; the useful structure is the two-dimensional geometry spanned by the uniform state and the marked subspace. The corresponding black-box improvement is quadratic and asymptotically optimal [4, 16].

The same interference mechanism has classical coherent-wave analogues, but that fact is not an efficient classical simulation of the encoded quantum algorithm. A wave realization may pay explicitly for the full lattice or mode structure, whereas a quantum implementation may encode the walk space more economically when the structure permits. Interference is therefore the positive mechanism, not by itself an account of computational advantage. An advantage requires end-to-end economy in preparation, oracle access, encoding, transformation, and readout, typically under a promise or oracle model.

V. UNCOMPUTATION, INFORMATION TRANSFER, AND RESET

A second misconception concerns garbage. Closed-system evolution is unitary,

$$|\psi\rangle \mapsto U|\psi\rangle, \quad U^\dagger U = I.$$

A non-injective classical function therefore cannot be implemented by a unitary acting only on the input register. It must be embedded reversibly, for example as

$$U_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle.$$

The information distinguishing the inputs remains somewhere in the global state.

If a reversible computation produces

$$|x\rangle_A |0\rangle_B |0\rangle_C \mapsto |x\rangle_A |f(x)\rangle_B |g(x)\rangle_C,$$

the garbage register C may be returned to $|0\rangle_C$ by applying the inverse computation to the retained correlated registers. This is local logical reset by uncomputation, not destruction of the global information.

No reset without information transfer. Let $|\phi_1\rangle_B$ and $|\phi_2\rangle_B$ be normalized states and suppose a unitary on B and an environment E satisfies

$$U_{BE} |\phi_i\rangle_B |0\rangle_E = |0\rangle_B |e_i\rangle_E, \quad i = 1, 2. \quad (7)$$

Preservation of inner products gives

$$\langle \phi_1 | \phi_2 \rangle = \langle e_1 | e_2 \rangle. \quad (8)$$

Thus Eq. (7) shows that resetting B preserves the distinguishing information in E . If the environment were also required to finish in one fixed state, $|e_1\rangle = |e_2\rangle$, then the two inputs would have to coincide as rays. An isometry can reset a subsystem by transferring information, but it cannot merge distinct global alternatives.

Consequently, if $|g(x)\rangle_B$ encodes a nonconstant classical function in orthogonal basis states, no unitary acting on B alone resets it for every x . The register can instead be handled by (i) applying the inverse computation to the retained correlated registers; (ii) transferring the information to an ancillary system or environment; or (iii) measuring it and retaining a classical record. In the latter two cases, any receiving memory that is to be reused must itself eventually be reset.

Bennett’s reversible-computation analysis shows how intermediate garbage can be uncomputed [17–19]. Thermodynamic erasure is a separate physical question. Resetting an unknown memory drawn from a nontrivial ensemble for cyclic reuse generally requires entropy export, but the minimum work or heat depends on the ensemble, bath, protocol, and available side information [20, 21]. A known standard state does not incur a universal $k_B T \ln 2$ cost merely because it occupies one bit of hardware. Whether the Landauer–Bennett account fully resolves Maxwell-demon arguments remains a philosophical dispute [22–24]; the operational statement used here is only that unitary dynamics preserves information globally.

VI. COPYABILITY IS BASIS-DEPENDENT

Classical information can be copied because it is encoded in distinguishable states. Quantum information cannot generally be copied. If a unitary cloner existed with

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle \quad \text{and} \quad U |\phi\rangle |0\rangle = |\phi\rangle |\phi\rangle,$$

then unitarity preserves inner products: on the one side,

$$\langle \psi | \langle 0 | U^\dagger U | \phi \rangle | 0 \rangle = \langle \psi | \phi \rangle \langle 0 | 0 \rangle = \langle \psi | \phi \rangle,$$

while on the other side the cloned outputs give

$$\langle \psi \psi | \phi \phi \rangle = \langle \psi | \phi \rangle^2.$$

Equating the inner products and taking absolute values gives

$$|\langle \psi | \phi \rangle| = |\langle \psi | \phi \rangle|^2.$$

Hence distinct clonable pure states are orthogonal; the alternative $|\langle \psi | \phi \rangle| = 1$ means that the normalized states coincide as rays. This is the no-cloning theorem [25, 26]. For mixed states, the no-broadcasting theorem says that a set of density operators can be broadcast if and only if they pairwise commute [27]. The no-deletion theorem gives a related obstruction: an unknown state cannot

be deleted from one of two copies while the other is left intact under the stated closed-system conditions [28].

These results identify a precise boundary of copyability. Within a specified orthonormal basis one may copy basis labels,

$$|x\rangle|0\rangle \mapsto |x\rangle|x\rangle,$$

but this does not clone an arbitrary superposition of those states. More generally, perfectly distinguishable classical labels can be redundantly recorded, and broadcastable mixed-state families commute.

Here a *context* means a maximal compatible set of observables, equivalently a maximal Boolean subalgebra of the projection lattice. In the nondegenerate finite-dimensional case it corresponds to an orthonormal basis and the projectors generated by it. A merely compatible subset need not be maximal.

Quantum computation can therefore manipulate information that is not globally copyable. A specified measurement instrument produces an outcome record in a chosen readout algebra. Environmental decoherence can select stable pointer information and suppress interference, but by itself it should not be identified with the production of one unique outcome in every interpretation.

VII. CONTEXTUALITY IS NOT MERE BASIS DEPENDENCE

Quantum algorithms use computational, phase, Fourier, and other bases, but a change of representation is not by itself a physical effect, and applying a Hadamard or Fourier transform is a unitary operation, not by itself a demonstration of contextuality. Kochen–Specker contextuality concerns a stronger obstruction involving compatible measurement contexts.

In Hilbert spaces of dimension at least three, the Kochen–Specker theorem rules out a global noncontextual value assignment satisfying the spectral and functional constraints—for example, assigning exactly one value 1 to every complete orthogonal resolution of the identity—independently of the other compatible observables measured alongside it [29–32]. This does not rule out contextual hidden-variable models.

For a single qubit, the standard Kochen–Specker obstruction for projective measurements is absent. Broader operational notions extend contextuality to preparations, transformations, and unsharp measurements [33]. The present partition formalism does not depend on resolving that terminological dispute.

Each partition observable A_Π in Eq. (5) is an ordinary Hermitian operator. Different partition observables may fail to commute, so their labels need not be jointly accessible on one system. That complementarity constrains complete identification, but it becomes a Kochen–Specker claim only when a specified collection of contexts satisfies the theorem’s global nonassignment conditions.

This separates two questions. Partition-resolving recombination explains how the accessible answer is produced; a resource theory asks why the same input–output statistics resist efficient classical simulation. Stabilizer operations admit efficient classical simulation under the Gottesman–Knill assumptions [34, 35]; universal quantum computation in the stabilizer/magic-state model therefore requires a resource outside that fragment. Depending on the model and dimension, nonstabilizer resources may be described by magic, quasiprobability negativity, or contextuality [36–39]. The cleanest equivalences occur in specified odd-prime-dimensional qudit settings, while qubits require additional qualifications. Contextuality is therefore neither asserted nor needed here as a necessary-and-sufficient explanation of every quantum speedup.

VIII. QUANTUM UNIVERSALITY VERSUS TURING UNIVERSALITY

The word “universal” has related but distinct uses in quantum information and in the theory of computation. A finite local gate set, together with the ability to apply its gates to specified qubits subject to the architecture, is approximately universal when, for every register size n , every target unitary U on that register, and every accuracy $\epsilon > 0$, a finite circuit over the gate set approximates U up to the physically irrelevant global phase. This quantifies over all n ; for a fixed n , the resulting circuits densely generate the appropriate unitary group. Density is an approximation statement. Compilation cost also depends on how the target is specified, and a generic n -qubit unitary has an exponentially large description [40].

Turing universality concerns effective symbolic procedures on finite inputs of unbounded length. A Turing-universal model can simulate any effective procedure under a finite encoding [41, 42]. This is the setting for partial recursive functions, undecidability, unbounded work space and time, and the halting problem.

A single circuit on n qubits implements one finite-dimensional map

$$U_n : \mathbb{C}^{2^n} \longrightarrow \mathbb{C}^{2^n}.$$

It has fixed width and depth. A circuit family $\{C_n\}_{n \geq 1}$ may nevertheless define a language even when it is nonuniform; such a family can hide arbitrary advice in the choice of its circuits. It constitutes an effective algorithmic model when a uniform rule generates C_n from n . For complexity-theoretic claims this rule is normally required to obey an explicit resource bound, such as polynomial time or logarithmic space.

With standard uniformity and error conventions, uniform quantum circuit families and quantum Turing machines yield equivalent bounded-error or Monte Carlo complexity classes. Exact and Las Vegas formulations require additional qualifications [7, 43, 44]. Thus the two models are positively related by simulation theorems, but

that equivalence is not supplied merely by the existence of a universal gate set.

For finite classical outputs, an effective halting condition, computable gate or transition amplitudes, and a fixed bounded-error gap, standard quantum models do not decide nonrecursive languages. Their possible advantage is one of complexity, not an enlargement of classical computability. Allowing noncomputable parameters or nonuniform noncomputable advice would change the model by inserting the desired information into its specification.

Finally, a Turing machine need not be modeled as repeatedly entering one many-to-one absorbing state: its transition function may simply stop being applied. The narrower unitary observation remains valid. If $U|h\rangle = |h\rangle$ and $U|\psi\rangle = |h\rangle$, injectivity implies $|\psi\rangle = |h\rangle$. Closed unitary dynamics therefore cannot merge distinct histories into one absorbing state, but standard quantum models implement operational termination with halt flags, clocks, external control, or measurement protocols [7, 43, 45, 46]. There is no conflict with quantum Turing universality.

IX. CONCLUSION

Quantum computation does not make the values associated with superposition components independently readable. It can instead encode a problem into a partition of its instance set and use coherent dynamics to map the cells of that partition to orthogonal output subspaces. The resulting answer can be resolved by the spectral projectors of a Hermitian observable or, equivalently, by a unitary that writes the cell label to a standard readout register.

This formulation also preserves the possibility of complete identification. A separating family of partitions has the singleton partition as its common refinement. If the corresponding observables can be jointly or sequentially implemented with the required resources, their labels identify the complete instance. Alternatively, a single enlarged unitary may resolve the refined label directly. Quantum algorithms need not be restricted to permanently coarse answers; the issue is whether the required partition or refinement admits an efficient physical realization. Parity makes this caveat vivid: even a two-cell partition can be hard to resolve when its cells are globally interlaced and the oracle exposes their label only through a high-degree dependence on all input coordinates.

The Bernstein–Vazirani leakage model makes the distinction between evaluation and resolution explicit. The answer function is evaluated by a globally unitary interaction even when a witness records path information. After the witness becomes inaccessible, the reduced channel loses selected coherences and therefore loses the corresponding partition-resolving interference. Orthogonal witness records produce complete dephasing of the selected coordinates, while nonorthogonal records give a

continuous visibility parameter. What is lost is accessible recombination, not global reversibility.

The broader constraints must remain separate. A subsystem can be reset unitarily only by retaining its distinguishing information elsewhere; thermodynamic erasure concerns the later physical reset of that information. Unknown states are not generally clonable. Basis changes are not by themselves Kochen–Specker contextuality, although specified collections of partition observables can instantiate contextuality scenarios. Finally, universal gate sets and Turing-universal uniform models are connected by simulation results but are not identical definitions.

Partition-resolving interference is therefore an operational framework, not a universal sufficient condition for speedup. Sampling tasks, quantum-state outputs, and other computational models may require a different immediate description. Within the oracle and identification problems considered here, however, the framework states the central point precisely: useful information is engineered into resolvable spectral alternatives, not recovered by branchwise inspection.

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DATA AVAILABILITY

No data were created or analyzed in this study.

Appendix A: A unitary leakage knob for Bernstein–Vazirani interference

The Bernstein–Vazirani algorithm gives a particularly transparent example of quantum speedup by interference rather than by any literal parallel readout of classical function values [6, 43, 47]. Let $a \in \mathbb{F}_2^n$ be an unknown string and define

$$f_a(x) = a \cdot x \pmod{2}, \quad x \in \mathbb{F}_2^n. \quad (\text{A1})$$

The usual reversible oracle is

$$O_a |x\rangle |y\rangle = |x\rangle |y \oplus f_a(x)\rangle. \quad (\text{A2})$$

With the target register initialized in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, this oracle acts by phase kickback,

$$O_a |x\rangle |-\rangle = (-1)^{a \cdot x} |x\rangle |-\rangle.$$

Thus

$$\begin{aligned} |0^n\rangle |-\rangle &\xrightarrow{H^{\otimes n}} 2^{-n/2} \sum_x |x\rangle |-\rangle \\ &\xrightarrow{O_a} 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle |-\rangle \\ &\xrightarrow{H^{\otimes n}} |a\rangle |-\rangle. \end{aligned} \quad (\text{A3})$$

Equivalently, the final amplitude of $|z\rangle$ is

$$2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{(a \oplus z) \cdot x} = \delta_{z,a}. \quad (\text{A4})$$

The hidden string is therefore recovered exactly with one coherent query. The essential step is not the evaluation of many classical values, but the conversion of a global phase character into a computational-basis population by a Fourier transform over $\mathbb{F}_2^n$.

We now introduce a reversible modification that degrades this interference without changing the Boolean function f_a . Let $S \subseteq \{1, \dots, n\}$ have cardinality k , and let $\bar{S}$ denote its complement. For $x \in \mathbb{F}_2^n$, write

$$x_S = (x_i)_{i \in S} \in \mathbb{F}_2^k, \quad x_{\bar{S}} = (x_i)_{i \in \bar{S}} \in \mathbb{F}_2^{n-k},$$

where a fixed ordering of the coordinates in S and $\bar{S}$ is understood. Thus x_S and $x_{\bar{S}}$ denote the subvectors of x supported on S and on its complement, respectively. Define the unitary permutation

$$W_a^S |x\rangle |y\rangle |e\rangle = |x\rangle |y \oplus f_a(x)\rangle |e \oplus x_S\rangle, \quad (\text{A5})$$

where the third register is a witness, garbage, or environmental register. The oracle still computes the same function f_a , and the enlarged map is reversible. However, if the third register is not coherently available to the algorithm, then it carries which-path information about the computational-basis label x . Here “which-path” is meant in the standard interferometric sense: a record relative to a chosen computational basis, not an independently readable classical computation carried by a component of the superposition. In this respect the construction is a tunable version of the familiar which-way suppression of interference [48–51]. It is also closely related to Aaronson’s Problem 11 on standard versus erasing oracles with input-dependent garbage, where sufficiently informative garbage is suggested to behave, under appropriate conditions, like decoherence or measurement of the response [52].

To specify the accessible query model, each call below receives a fresh witness register in $|0^k\rangle$, and that register is discarded after the interaction. Thus the algorithm has access to the memoryless channel

$$\Phi_a^S(\rho) = \text{Tr}_E [W_a^S(\rho \otimes |0^k\rangle\langle 0^k|_E)(W_a^S)^\dagger] \quad (\text{A6})$$

on the query and answer registers, not to the inverse of the enlarged unitary. This distinction fixes the oracle interface for repeated calls. For a computational-basis value query $|x\rangle|0\rangle$, discarding the witness does not obscure the definite answer $f_a(x)$; it suppresses only coherences between inputs carrying distinguishable witness records. Accordingly, this is structured coordinate dephasing, not generic depolarizing noise: it preserves the within-block coherences that resolve $a_{\bar{S}}$, completely removes selected cross-block coherences when the records are orthogonal, and leaves ordinary basis-state value queries intact.

For later query-complexity statements, the phase-kickback restriction of the accessible channel is useful to record explicitly:

$$\mathcal{E}_a^S(\rho) = D_a \left(\sum_{s \in \mathbb{F}_2^k} P_s \rho P_s \right) D_a^\dagger, \quad (\text{A7})$$

where

$$D_a = \sum_x (-1)^{a \cdot x} |x\rangle \langle x|, \quad P_s = \sum_{x: x_S = s} |x\rangle \langle x|. \quad (\text{A8})$$

Equivalently, in matrix elements,

$$\mathcal{E}_a^S(\rho) = \sum_{\substack{x, x' \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} \rho_{xx'} |x\rangle \langle x'|. \quad (\text{A9})$$

This is the phase-interference restriction, not the full accessible query–answer channel Φ_a^S : the latter still depends on a_S through the answer-register flip, which is why ordinary value queries can recover the hidden components.

Proposition: partial loss of the Bernstein–Vazirani interference pattern. Let the query register be initialized in $|+\rangle^{\otimes n}$, the answer register in $|-\rangle$, and the witness register in $|0^k\rangle$. Apply W_a^S , trace out the witness register, and then apply $H^{\otimes n}$ to the query register. The final measurement distribution on the query register is

$$\Pr(z) = 2^{-k} \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \quad (\text{A10})$$

Thus one coherent Bernstein–Vazirani-style query determines the $n - k$ components of a outside S with certainty, while the observed coordinates z_S are uniformly distributed over $\mathbb{F}_2^k$ and carry no information about a_S . In particular, the usual one-query speedup survives only on those coordinates whose computational-basis path labels have not been exported to the inaccessible witness.

In the language of Sec. III, this measurement exactly resolves the partition

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \quad C_u = \{a \in \mathbb{F}_2^n : a_{\bar{S}} = u\}. \quad (\text{A11})$$

Indeed, after a fixed reordering of output coordinates, the Hermitian observable

$$A_{\Pi_S} = \sum_{u \in \mathbb{F}_2^{n-k}} \lambda_u (|u\rangle\langle u|_{\bar{S}} \otimes I_S), \quad \lambda_u \neq \lambda_v \text{ for } u \neq v, \quad (\text{A12})$$

has spectral projectors labelled by the cells of Π_S . The remaining value queries refine this partition to the singleton partition and hence complete the identification of a .

Operationally, after this first query one may recover the remaining coordinates by k ordinary value queries on the standard basis vectors e_i , $i \in S$, since $f_a(e_i) = a_i$. This provides an explicit completion strategy with query count $1 + k$ for $k < n$. At the endpoint $k = n$, the first Bernstein–Vazirani-style query gives the uniform distribution and is best omitted, so the standard n value queries suffice. Thus the interference step and ordinary value recovery are not opposed: a structure-matched interference readout can supply the part of the answer that remains coherently recombinable, while additional value queries can complete the full classical string. Hence this simple strategy interpolates as

$$Q_{\text{strat}}(n, k) = \begin{cases} 1 + k, & 0 \leq k < n, \\ n, & k = n. \end{cases} \quad (\text{A13})$$

Equation (A13) describes an explicit completion strategy, not a general optimality theorem for arbitrary adaptive use of the full leaked channel. The structural fact proved below is narrower: the accessible phase-interference channel contains no dependence on a_S . Once $a_{\bar{S}}$ is known, the k standard-basis value queries used here provide k independent linear equations and are necessary within this restricted second-stage strategy. Optimal query complexity for arbitrary adaptive protocols using the full query–answer channel is a separate problem and is not claimed here.

Proof. After the first Hadamard transform and the application of W_a^S , the joint state is

$$|\psi_a^S\rangle = 2^{-n/2} \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x} |x\rangle |-\rangle |x_S\rangle. \quad (\text{A14})$$

Tracing out the witness register gives the reduced density operator on the query register

$$\rho_a^S = 2^{-n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} |x\rangle\langle x'|. \quad (\text{A15})$$

Hence coherences remain only between computational-basis components that agree on the coordinates in S .

The witness register has therefore implemented a perfect which-path record for those coordinates.

Now apply $H^{\otimes n}$ and measure in the computational basis. For $z \in \mathbb{F}_2^n$,

$$\Pr(z) = \langle z | H^{\otimes n} \rho_a^S H^{\otimes n} | z \rangle \quad (\text{A16})$$

$$= 2^{-2n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x') + z \cdot (x \oplus x')}. \quad (\text{A17})$$

Set $t = x \oplus x'$. The constraint $x_S = x'_S$ is equivalent to $t_S = 0$. For each such t , there are 2^n choices of x . Therefore

$$\Pr(z) = 2^{-n} \sum_{\substack{t \in \mathbb{F}_2^n \\ t_S = 0}} (-1)^{(a \oplus z) \cdot t} \quad (\text{A18})$$

$$= 2^{-n} \sum_{t_{\bar{S}} \in \mathbb{F}_2^{n-k}} (-1)^{(a_{\bar{S}} \oplus z_{\bar{S}}) \cdot t_{\bar{S}}}. \quad (\text{A19})$$

By character orthogonality over $\mathbb{F}_2^{n-k}$,

$$\sum_{t_{\bar{S}} \in \mathbb{F}_2^{n-k}} (-1)^{(a_{\bar{S}} \oplus z_{\bar{S}}) \cdot t_{\bar{S}}} = \begin{cases} 2^{n-k}, & z_{\bar{S}} = a_{\bar{S}}, \\ 0, & z_{\bar{S}} \neq a_{\bar{S}}. \end{cases} \quad (\text{A20})$$

This proves Eq. (A10).

The block form above identifies what information remains available through the phase-interference channel. Inside each block $x_S = x'_S = s$, the contribution of a_S cancels:

$$a \cdot (x \oplus x') = a_{\bar{S}} \cdot (x_{\bar{S}} \oplus x'_{\bar{S}}). \quad (\text{A21})$$

Therefore the reduced phase-interference channel depends on a only through $a_{\bar{S}}$. The hidden components a_S are not merely hard to extract by the final Hadamard transform; they are absent from the accessible phase channel once the witness has been traced out.

The same dephasing in the x_S basis occurs for any preparation of the answer register, since the witness records x_S independently of the answer-register state. The dephasing removes the dependence on a_S from the accessible *phase-interference* channel; it does not remove ordinary value information carried by the answer register. Consequently, once $a_{\bar{S}}$ has been obtained by the surviving Bernstein–Vazirani interference, the remaining components a_S can be recovered by standard value-query use of

$$f_a(x) = a_S \cdot x_S + a_{\bar{S}} \cdot x_{\bar{S}}. \quad (\text{A22})$$

Since $a_{\bar{S}}$ is then known, a standard value query supplies one linear equation over the unknown vector $a_S \in \mathbb{F}_2^k$. Querying the standard basis vectors e_i , $i \in S$, recovers these k components directly. This proves the stated interpolation strategy in Eq. (A13). $\square$

1. Continuous witness overlap

The preceding construction has a literal continuous version. For every $i \in S$, choose normalized witness states $|\epsilon_0^{(i)}\rangle$ and $|\epsilon_1^{(i)}\rangle$ with real overlap

$$\langle \epsilon_0^{(i)} | \epsilon_1^{(i)} \rangle = \lambda_i, \quad 0 \leq \lambda_i \leq 1, \quad (\text{A23})$$

and put $|\epsilon_{x_S}\rangle = \bigotimes_{i \in S} |\epsilon_{x_i}^{(i)}\rangle$. A controlled isometry, extendable to a unitary on a sufficiently large witness space, can act on its initialized subspace as

$$\widetilde{W}_a^{S,\lambda} |x\rangle |y\rangle |0\rangle_E = |x\rangle |y \oplus f_a(x)\rangle |\epsilon_{x_S}\rangle. \quad (\text{A24})$$

The retained query label x makes the images of distinct computational basis inputs orthogonal, so nonorthogonality of the witness states does not conflict with this unitary extension.

After phase kickback and tracing out the witness, the query-register matrix elements are

$$\rho_{xx'} = 2^{-n} (-1)^{a \cdot (x \oplus x')} \prod_{i \in S} \lambda_i^{x_i \oplus x'_i}. \quad (\text{A25})$$

The final Hadamard transform therefore gives the explicit distribution

$$\Pr_{\lambda}(z) = \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}] \prod_{i \in S} \frac{1 + (-1)^{z_i \oplus a_i} \lambda_i}{2}. \quad (\text{A26})$$

One way to see the factorization is to note that each leaked coordinate has a reduced two-dimensional state whose off-diagonal coherence is multiplied by λ_i ; the last Hadamard then returns the correct bit with probability $(1 + \lambda_i)/2$. In particular,

$$\Pr_{\lambda}(z = a) = \prod_{i \in S} \frac{1 + \lambda_i}{2}. \quad (\text{A27})$$

The endpoint $\lambda_i = 1$ leaves that coordinate fully coherent, whereas $\lambda_i = 0$ makes the two witness records orthogonal and yields the uniform factor in Eq. (A10). Intermediate overlaps thus continuously interpolate the visibility of the corresponding partition-resolving interference. Establishing an optimal adaptive query complexity for every overlap vector is a separate problem; the formula above characterizes the stated one-query readout

exactly. If $\lambda_i = 1$ for only some $i \in S$, those coordinates are effectively not leaked and the corresponding factors reduce to $\mathbf{1}[z_i = a_i]$; the resolved partition is then finer than the partition associated with S alone.

The two extremes make the physical point explicit. For $k = 0$, no computational-basis path label is exported and the ordinary one-query Bernstein–Vazirani interference pattern is recovered. For $k = n$, the witness records the entire computational-basis label x , all off-diagonal terms in that basis are removed from the algorithm’s accessible state, and the final Hadamard transform has no global phase pattern left to recombine. The Boolean function f_a has still been evaluated coherently by a unitary permutation on the enlarged system. What has been lost is not evaluation but recombination.

This example shows that reversibility alone is not the relevant criterion. The map W_a^S is a unitary permutation on the enlarged Hilbert space. If the witness register remains coherently accessible, one may uncompute it and restore the full interference pattern. If, however, the witness register has escaped into an inaccessible environment, then the reduced dynamics available to the algorithm is no longer reversible. The witness register is precisely the residual garbage identified in the main text as an obstruction to free unitary reset, here promoted to a tunable knob. The one-query Bernstein–Vazirani readout advantage is lost in this leakage model not because the Boolean function f_a has changed, nor because the global dynamics has ceased to be unitary, but because the inaccessible witness dephases the coherences needed for the later Fourier-basis recombination.

Pointedly stated, this demonstrates that a quantum computer does not “compute all branches at once and let us independently read them”—where the scare-quoted branch language names precisely the classical picture being rejected. If it did, leaking the computational-basis path labels to a witness register would not matter: the function $f_a(x)$ was still computed. Instead, the appendix shows that the speedup relies on *recombining amplitudes* by interference in the basis matched to the hidden linear character. When uncomputation is blocked by an inaccessible witness, the accessible dynamics is dephased in the computational basis, the interference is quenched, and the lost coordinates must be recovered, if at all, by ordinary value queries.

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